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TME–406

B. TECH. (ME) (FOURTH SEMESTER)
MID SEMESTER
EXAMINATION, March, 2024

MECHANICS OF MATERIAL

Time : 1½ Hours

Maximum Marks : 50

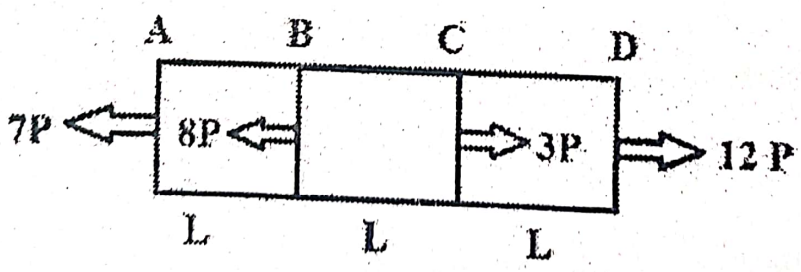
Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Find out the elongation in conical bar due to its self-weight, and draw the axial load diagram. (CO1)

OR

(b) Prismatic bar as shown in the figure find out the resultant change in the length of the bar and draw the axial load diagram. (CO1)



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2. (a) Write the expression for volumetric strain of rectangular bar subjected to uni-axial loading and prove that : (CO2)

volumetric strain = longitudinal strain + 2 × lateral strain.

OR

(b) Write the expression for stresses in oblique plane for prismatic bar under biaxial stress system and draw the Mohr's circle also. (CO2)

3. (a) Explain Hooke's law and principal stresses. (CO1)

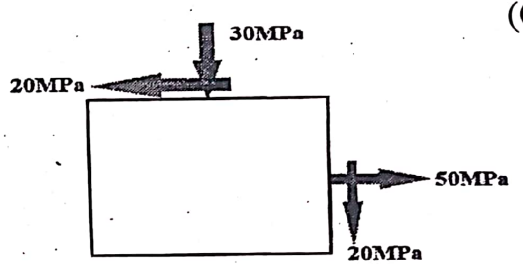
OR

(b) A rectangular bar of cross-section 50 mm × 20 mm is subjected to axial pull of 100 kN. Determine change in length, change in width, change in thickness. Assume L = 1 m, $E = 2 \times 10^5$ N/mm^2 and Poisson's ratio $\mu = 0.25$. (CO1)

4. (a) Explain stress-strain curve for low carbon steel with suitable diagram. (CO1)

OR

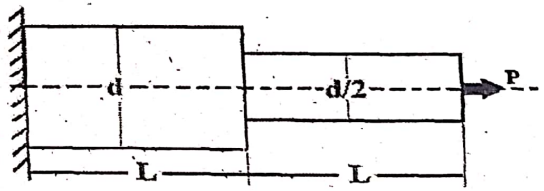
(b) Draw the Mohr's circle for below diagram. (CO1)



5. (a) Write about principle of superposition. What happen in the prismatic bar if it is loaded under axial load P and 2P which are acting opposite to each other at end of bar through longitudinal axis left and right side. (CO2)

OR

(b) Find out net elongation and ratio of maximum stress to minimum stress in below figure. (CO2)



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